

CMA File Watcher Service - Support Guide

Audience: New SalesOps or IT support staff responsible for CMA intake and PCF creation support.

Scope: How the service works, how to install/deploy it, and what inbound file structure is required for successful processing.

1) What This Program Does

The Windows service watches the inbound CMA share for new Excel files. For each file, it reads key header fields and item rows, writes records to BAT_App.dbo.Chap_Cmaltems, validates business rules, runs stored procedures to create PCFs, emails the submitter, archives the workbook, and removes the original from the inbound folder.

Main flow from import to PCF creation

- User submits CMA workbook to \ciiedi01\SendDocs\CMAInbound (file extension .xlsx).
- FileSystemWatcher detects file creation and enqueues it (single file processing queue).
- Service waits until the file is unlocked and non-empty.
- Service reads worksheet data (prefers sheet name "CMA Template", then "CMA", then first sheet).
- Service inserts one Chap_Cmaltems row per item line in the workbook.
- CMAValidator checks required fields, date rules, duplicate conflicts, and item validity.
- If valid, service runs CreatePcfFromChapCmaltems_sp (and sp_ArchiveReplacedPCF for replacements).
- If PCF number exists after procedure, success email is sent and file is archived to Processed.
- If validation or creation fails, failure email is sent (validation failures) and file is archived to Rejected.
- Original inbound workbook is deleted only after archive copy exists in Processed or Rejected.

Service dependencies and integration points

Component	Used For
Windows Service (CmaFileWatcherService.exe)	Continuous file monitoring and orchestration
Inbound Share (CMAInbound)	Receives CMA workbook files
SQL Server (ciisql10 BAT_App)	Staging records in Chap_Cmaltems and email dispatch via Database Mail
Linked server (CiiSQL01 + PCF DB)	Existing PCF conflict checks against ProgControl/pcitems
Stored Procedures	CreatePcfFromChapCmaltems_sp and sp_ArchiveReplacedPCF drive PCF lifecycle
Database Mail profile SalesOps	Sends success/failure notifications to submitter email

2) Installation and Deployment

Build prerequisites

- Windows host capable of running a .NET Framework 4.7.2 service.
- Visual Studio/MSBuild with NuGet restore support.
- Access to referenced package folder structure (../packages) from solution path.
- Network access and permissions for inbound share and SQL Server ciisql10.

Build steps (from repository root)

- Restore NuGet packages (if not already restored).
- Build solution CmaFileWatcherSvc.sln (Debug or Release).
- Verify output contains CmaFileWatcherService.exe and dependent DLLs in CmaFileWatcherService/bin/.

Deploy to service server (example from code comments)

- Copy compiled output to target folder (example: C:\CmaFileWatcher).
- Stop existing service: `sc stop CmaFileWatcherService`
- Delete existing registration: `sc delete CmaFileWatcherService`
- Create service: `sc create CmaFileWatcherService binPath= "C:\CmaFileWatcher\CmaFileWatcherService.exe" start= auto`
- Start service: `sc start CmaFileWatcherService`
- Check status: `sc query CmaFileWatcherService`

Runtime configuration and controls

Setting/Control	Behavior
config file path	\\ciiedi01\SendDocs\CMAInbound\config.ini
PcfDatabase (INI)	Used by validator when checking linked server PCF conflicts
Watch folder	Hardcoded to \\ciiedi01\SendDocs\CMAInbound in service code
Watched file type	.xlsx only
Debug logging switch	Set environment variable DebugCMAWatcher=true
Logs	CmaFileWatcherService.log always; debug.log only when debug switch enabled

Security and operational notes

- Current code contains direct SQL credentials in connection strings; treat binaries and source as sensitive.
- Service account must have read/write/create/delete rights on inbound, Processed, and Rejected folders.
- Service account must execute required SQL queries/stored procedures and send dbmail via profile SalesOps.
- If file remains locked longer than retry window, it is skipped and logged as inaccessible.

3) Required Inbound Workbook Structure (what support should validate)

Important clarification: this service processes Excel workbooks (.xlsx), not PDF files directly. In practice, users may generate this workbook from their CMA process/template, but the watched input is an Excel file.

Expected file naming

- Extension must be .xlsx.
- File name should include "_sentby_" so submitter email can be extracted.
- Accepted sender domains in filename parser: @chapinmfg.com or @chapinusa.com.
- Example: CUSTOMER123_sentby_user@chapinmfg.com.xlsx

Worksheet selection

- First choice: sheet named "CMA Template".
- Second choice: sheet named "CMA".
- Fallback: first worksheet in workbook.

Header fields the service reads

Field	Cell Logic Used by Service	Purpose
Existing PCF (replacement check)	A2	If numeric, request is treated as replacement CMA
PCF Type heading row	Search B1:B10 for "PCF Type"	Determines placement of promo fields
Start date anchor row	Search P1:P10 for first parseable date	Base row for customer/date offsets
Promo Terms	F5 (or F2 when start row is 1)	Required for PD/PW types
Promo Freight Terms	C5 (or C2 when start row is 1)	Required for PD/PW types
Promo Freight Minimums	D5 (or D2 when start row is 1)	Required for PD/PW types
Promo Freight Other Amount	D6 (or D3 when start row is 1)	Optional value captured to database
Customer/Corp fields	B row offsets from detected start row	Used to derive Cust_num, Corp_flag, archive name
Buying Group	K(startRow+2)	If value is #N/A it is converted to blank
Submitted By	M(startRow+2)	Stored in Chap_Cmaltems
Start Date	P(startRow)	Stored in Chap_Cmaltems.StartDate
End Date	P(startRow+2)	Stored in Chap_Cmaltems.EndDate
General Notes	If A1 is "General Notes", uses B1	Stored as GenNotes

Item detail rows

- Item parsing starts at row (startRow + 5).
- Loop continues while column B (Item #) is non-empty.
- Per row mapping: A=Description, B=Item #, D=Sell Price (non-numeric defaults to 0).
- One database row is inserted per item into Chap_Cmaltems with shared header values.

Example from sample workbook in repository

- Row 10 contains customer number and P-column label for End Date in the sample.
- Row 12 begins item grid headers (Description, Item #, cost/sell columns).
- Actual item records start at row 13 and continue down column B.

4) Validation Rules and Failure Behavior

Validation checks executed by CMAValidator

- Required checks on first record: PcfTypeText, StartDate, EndDate.
- For PcfType starting PD or PW: Promo Terms, Promo Freight Terms, and Promo Freight Minimums are required.
- Date integrity: StartDate cannot be after EndDate.
- Conflict check: same customer, same start date, same item cannot overlap an active PCF record (status 3, not 98), excluding the replaced PCF when applicable.
- Item check: each item must exist in item_mst and not be obsolete (stat != O).

On validation failure

- Rows for the CMA are marked Status = F when PCFNumber is null and status was N.
- Failure email sent to parsed submitter email with list of validation errors.
- Workbook copy saved under Rejected folder with generated archive name.

On success

- Optional replacement path archives old PCF first (sp_ArchiveReplacedPCF).
- PCF creation procedure executed (CreatePcfFromChapCmaltems_sp).
- If PCF number is populated, success email is sent and workbook saved under Processed.
- Service writes returned PCF number into cell A2 in archived workbook copy.

Archive naming behavior

- archiveName = __
- baseFilename removes "_sentby_" part but preserves extension.

5) Support Runbook: Fast Triage Checklist

Symptom	What to check first
No processing started	Service status, watch path availability, file extension is .xlsx, file not locked.
File processed but rejected	CmaFileWatcherService.log + failure email details + template required fields.
No success/failure email	Filename contains parseable sender email/domain and dbmail profile SalesOps health.
PCF not created	Stored procedure execution rights, SQL errors, Chap_Cmaltems rows and validation state.
Original file still in inbound	Whether archive copy exists; deletion happens only after Processed/Rejected write.

Key SQL objects referenced by this service

- Tables: Chap_Cmaltems, item_mst, linked ProgControl, linked pcitems.
- Procedures: CreatePcfFromChapCmaltems_sp, sp_ArchiveReplacedPCF.
- Mail proc: msdb.dbo.sp_send_dbmail (profile SalesOps).

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